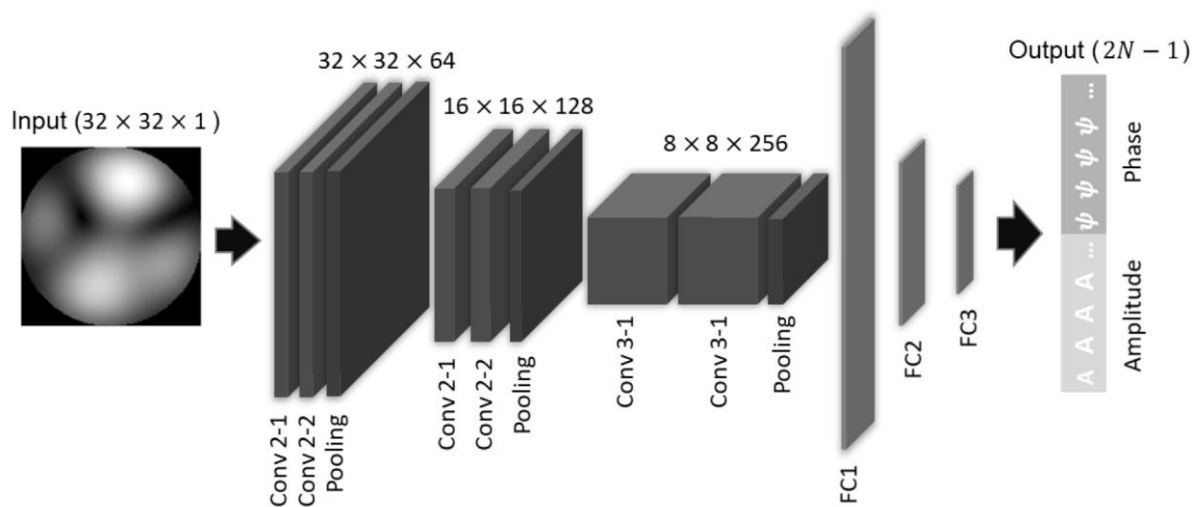


Protocol

Modal Decomposition On A Multimode Fiber



Task 1: Training A MLP For Mode Compensation For 3 Modes.

Hyperparameters:

MLP	Epochs	Learning rate	Mini batch size	Time (min)
3 modes	10	0.001	128	0.82

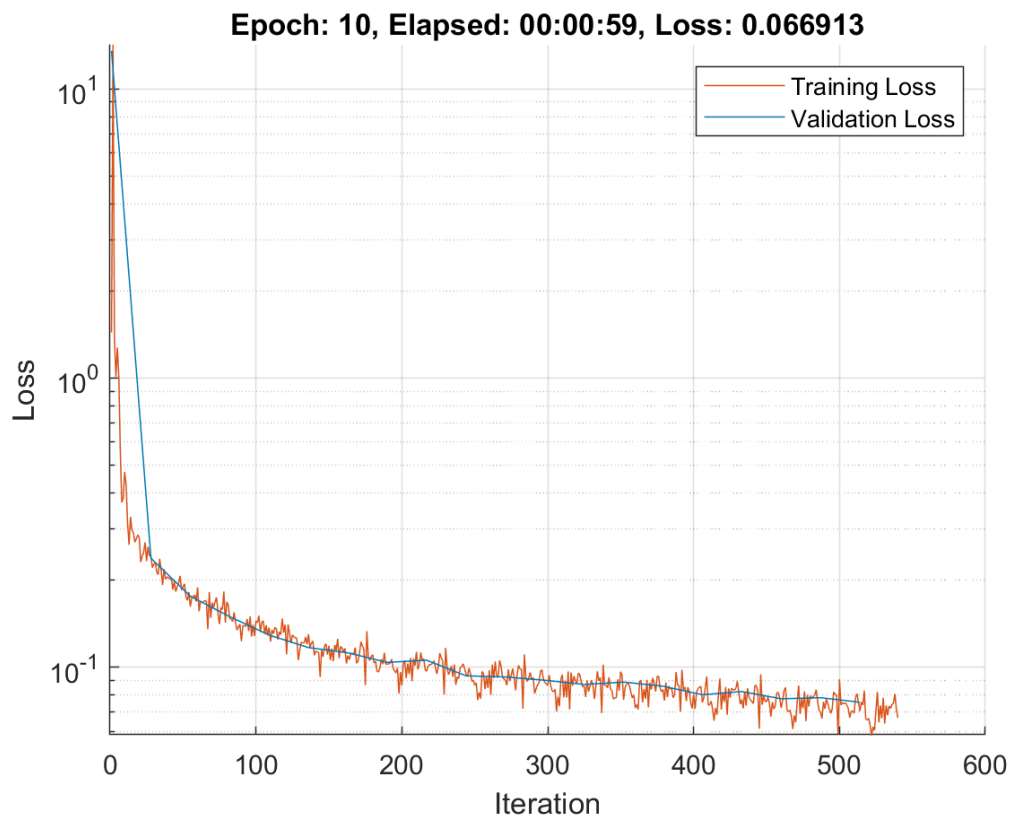


Figure 1: Training-Loss-Progression MLP, 3 Modes

Task 2: Evaluating The Trained MLP For 3 Modes.

The trained nets are evaluated on the test data. Results are visualized in a suitable plot using the cross correlation coefficient (Γ). The average ($\bar{\Gamma}$) as well as the standard deviation (σ) are calculated as well. Furthermore, the relative deviation of the prediction is calculated using data on amplitude (ρ) and phase (φ).

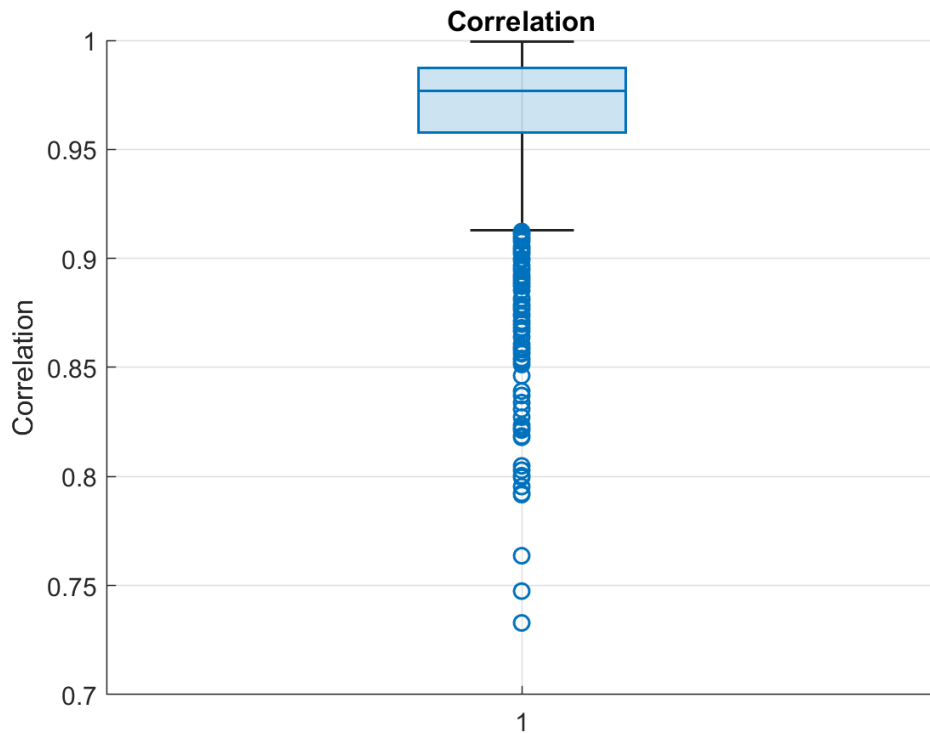


Figure 2: Evaluation MLP, 3 Modes

MLP	$\bar{\Gamma}$	σ	$\bar{\Delta\rho}$	$\bar{\Delta\varphi}$
3 modes	96.7%	3.243%	9.1%	24.1%

Description:

- The MLP reaches a high average Correlation $\bar{\Gamma}$ of 96.7%
- The standard deviation σ of the results is 3.243%.
- The results show a significant discrepancy between predictive accuracy in amplitude ($\bar{\Delta\rho} = 9.1\%$) and phase ($\bar{\Delta\varphi} = 24.1\%$), indicating that phase reconstruction is more complex than amplitude reconstruction.
- The boxplot (Figure 2) shows cross correlation values clearly accumulating above 0.95 with outliers down to approximately 0.73.

Task 3: Training And Evaluating A MLP For Mode Compensation For 5 Modes

Below, you may find the training progress plotted in the form of **TrainingLoss = $f(\text{iteration})$** . The trained networks are evaluated on the test data. Results are visualized in a plot (Figure 4) using the correlation coefficient (Γ). Furthermore, the average correlation coefficient ($\overline{\Gamma}$), the standard deviation (σ) as well as the relative deviation of the predicted weights for the amplitude (ρ) and phase (φ) are listed in the table below Figure 4.

Hyperparameters:

MLP	Epochs	Learning rate	Mini batch size	Time (min)
5 modes	10	0.001	128	0.733

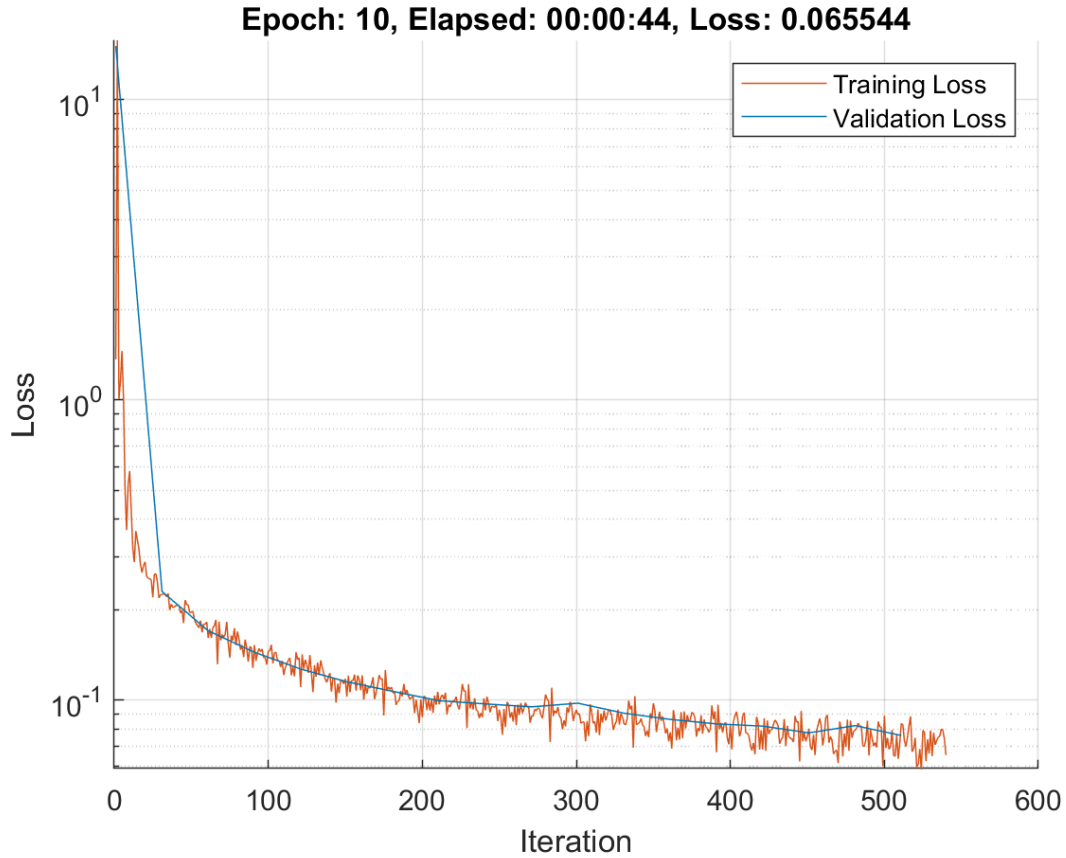


Figure 3: Training-Loss-Progression MLP, 5 Modes

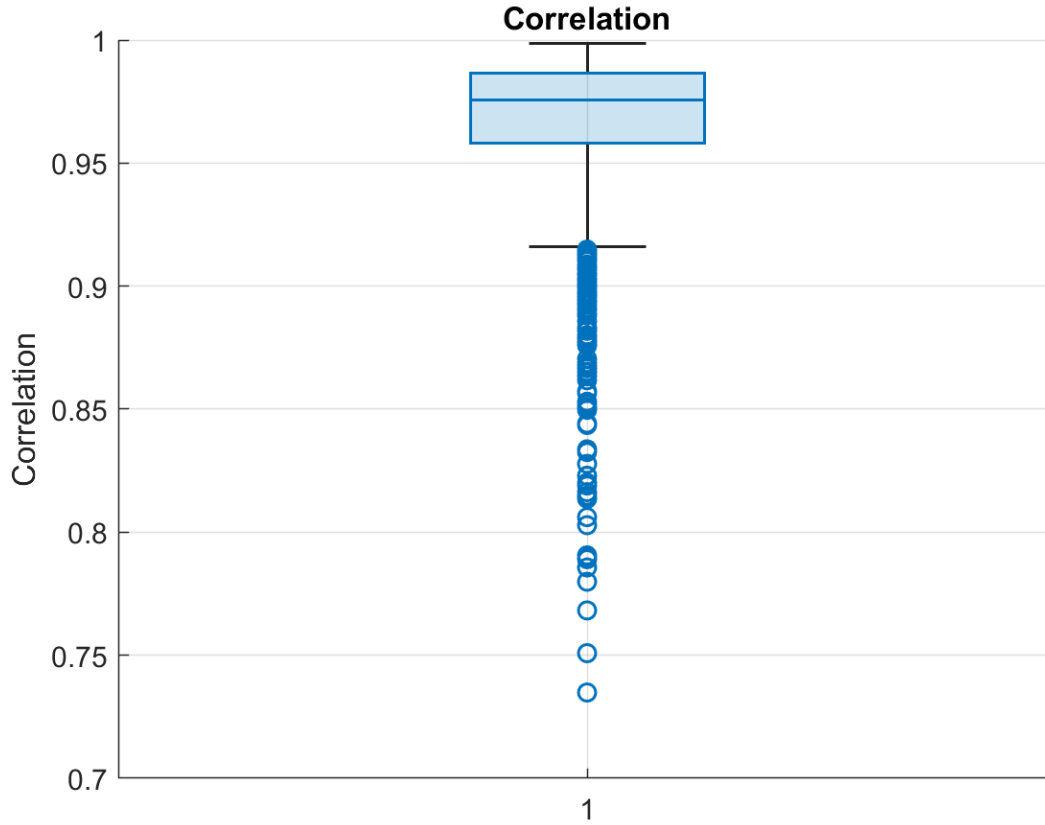


Figure 4: Evaluation MLP, 5 Modes

MLP	$\bar{\Gamma}$	σ	$\overline{\Delta\rho}$	$\overline{\Delta\varphi}$
5-modes	96.6%	3.303%	9.9%	24.7%

Description:

- Expanding to 5 modes does not seem to impact the average correlation $\bar{\Gamma} = 96.6\%$ significantly as it almost stays constant.
- Both the relative deviations in amplitude ($\overline{\Delta\rho} = 9.9\%$) as well as in phase ($\overline{\Delta\varphi} = 24.7\%$) are slightly higher when compared to the 3 mode case.
- The standard deviation is increased slightly to $\sigma = 3.303\%$.
- The results indicate that the MLP architecture is robust when it comes to increasing the number of modes, even at the cost of a slight decrease in accuracy.

Task 4: Training And Evaluating A VGG Network For Mode Compensation For 3 And 5 Modes

Training progress and results are plotted as in the previous tasks.

Hyperparameters:

VGG	Epochs	Learning rate	Mini batch size	Time (min)
3 modes	10	0.001	128	12.85
5 modes	10	0.001	128	23.4

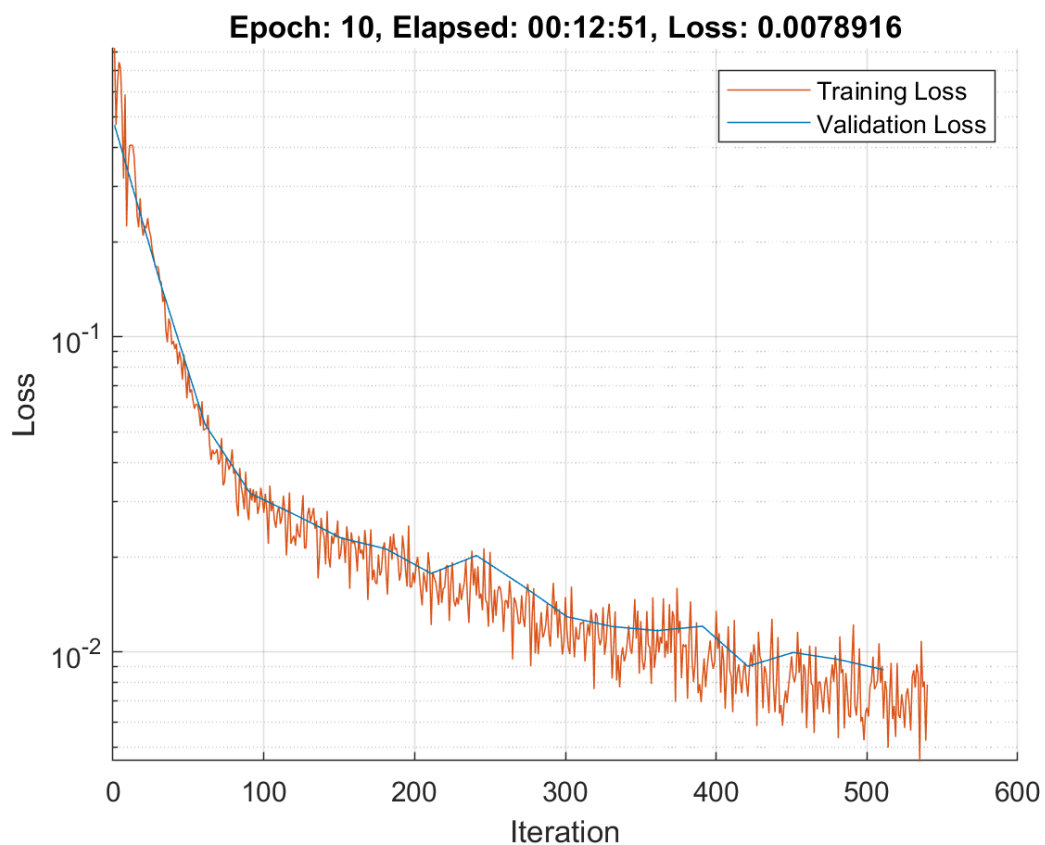


Figure 5: Training-Loss-Progression VGG, 3 Modes

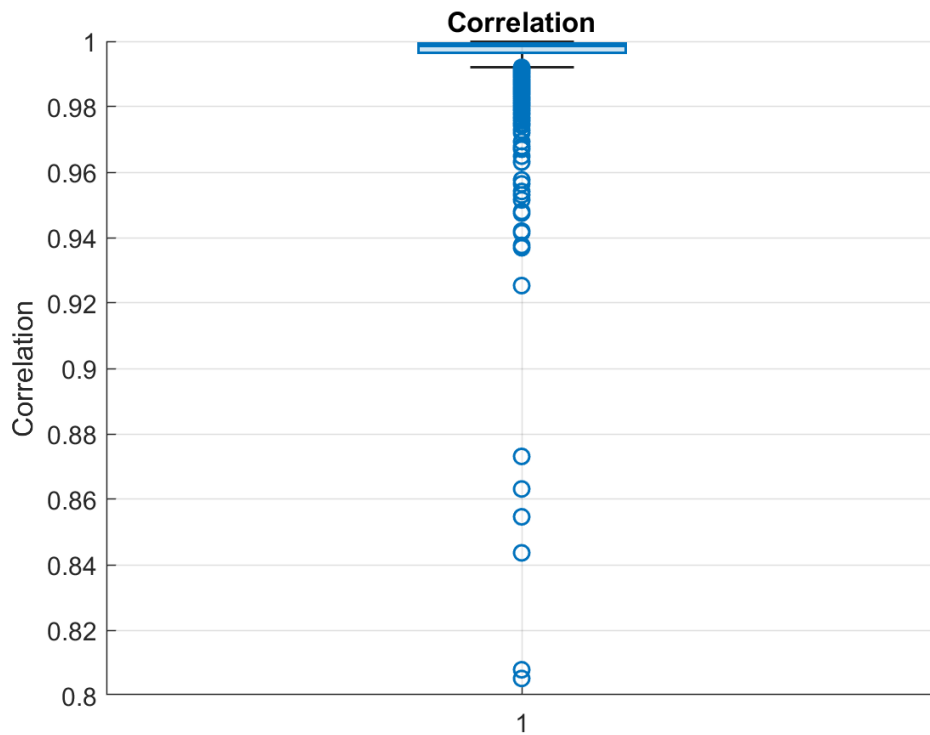


Figure 6: Evaluation VGG, 3 Modes

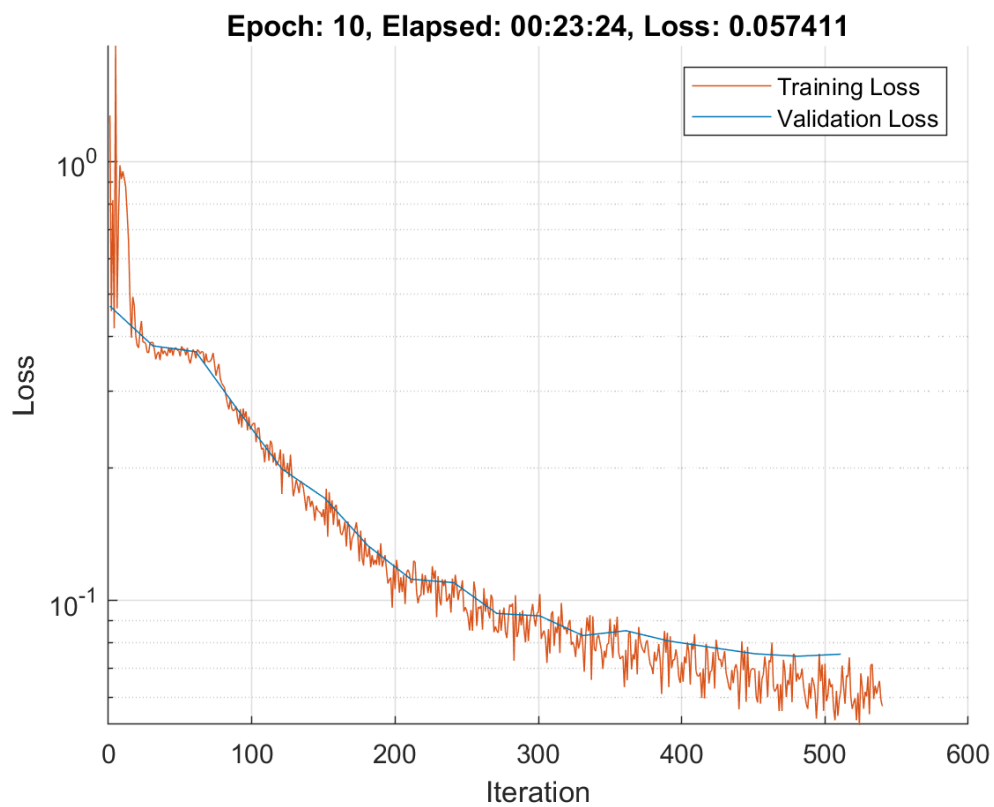


Figure 7: Training-Loss-Progression VGG, 5 Modes

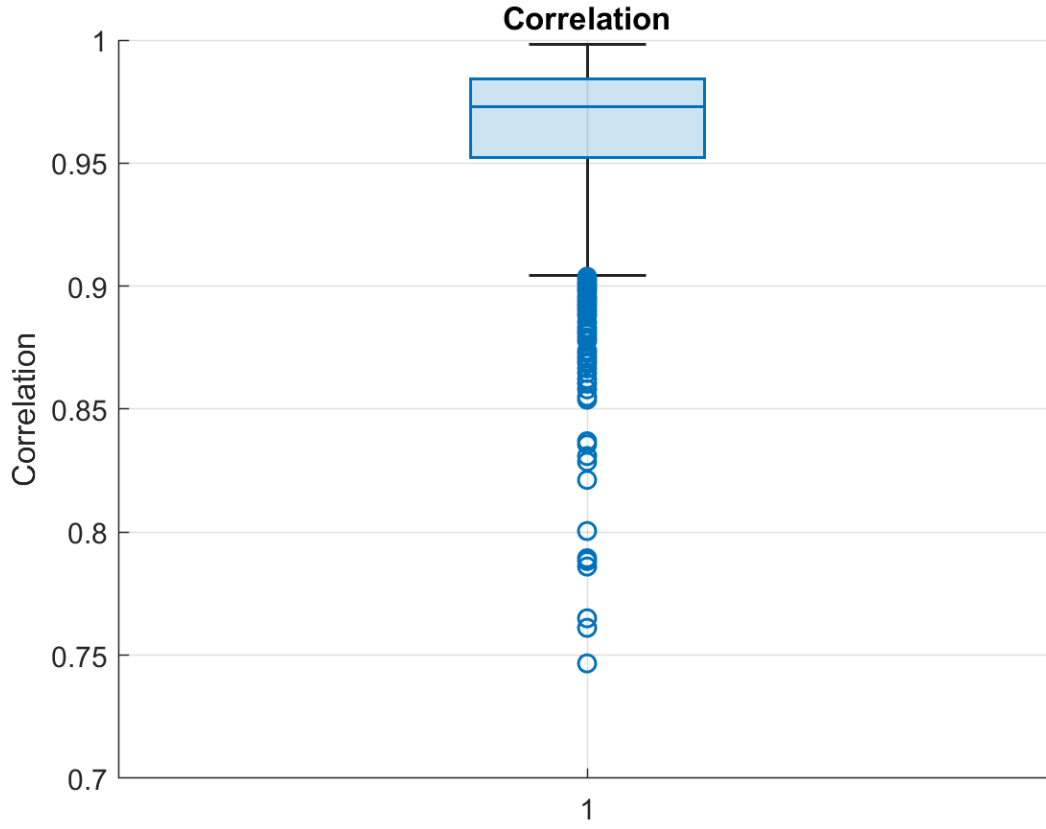


Figure 8: Evaluation VGG, 5 Modes

VGG	$\bar{\Gamma}$	σ	$\overline{\Delta\rho}$	$\overline{\Delta\varphi}$
3-mode	99.6%	1.207%	2.2%	6.5%
5-mode	96.4%	3.045%	8.4%	23.5%

Task 5: Training A VGG Network For Mode Compensation For 5 Modes Using Transfer Learning

A VGG is trained for 5 modes on 10,000 datapoints using transfer learning. The VGG is initialized using the learned model parameters of the 3 mode VGG from task 4. The results are then compared to those for the 5 mode VGG from task 4. Finally, training is continued on the training data with 50,000 datapoints. The results are visualized.

Hyperparameters:

VGG	Training data	Epochs	Learning rate	Mini batch size	Time (min)
5-mode-TL	10000	10	0.001	128	15.6
5-mode-TL	50000	10	0.001	128	

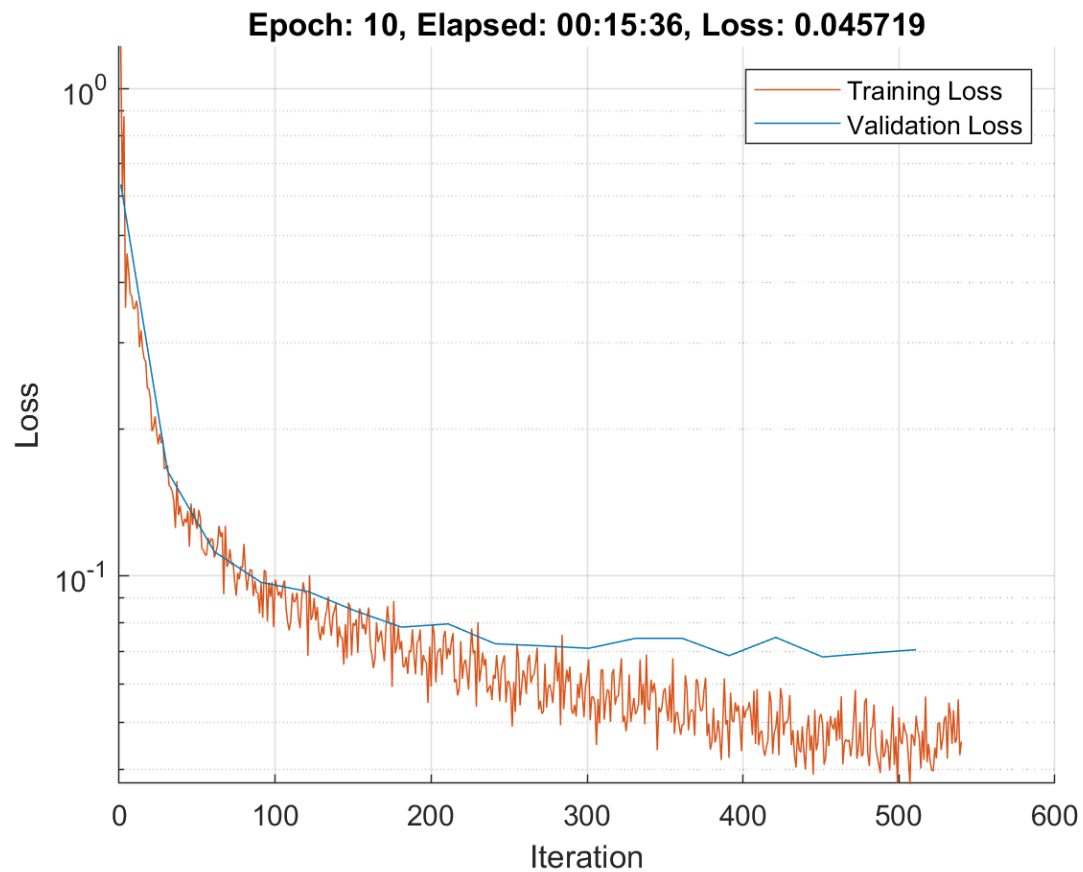


Figure 9: Training-Loss-Progression VGG TL, 5 Modes, 10,000 Datapoints

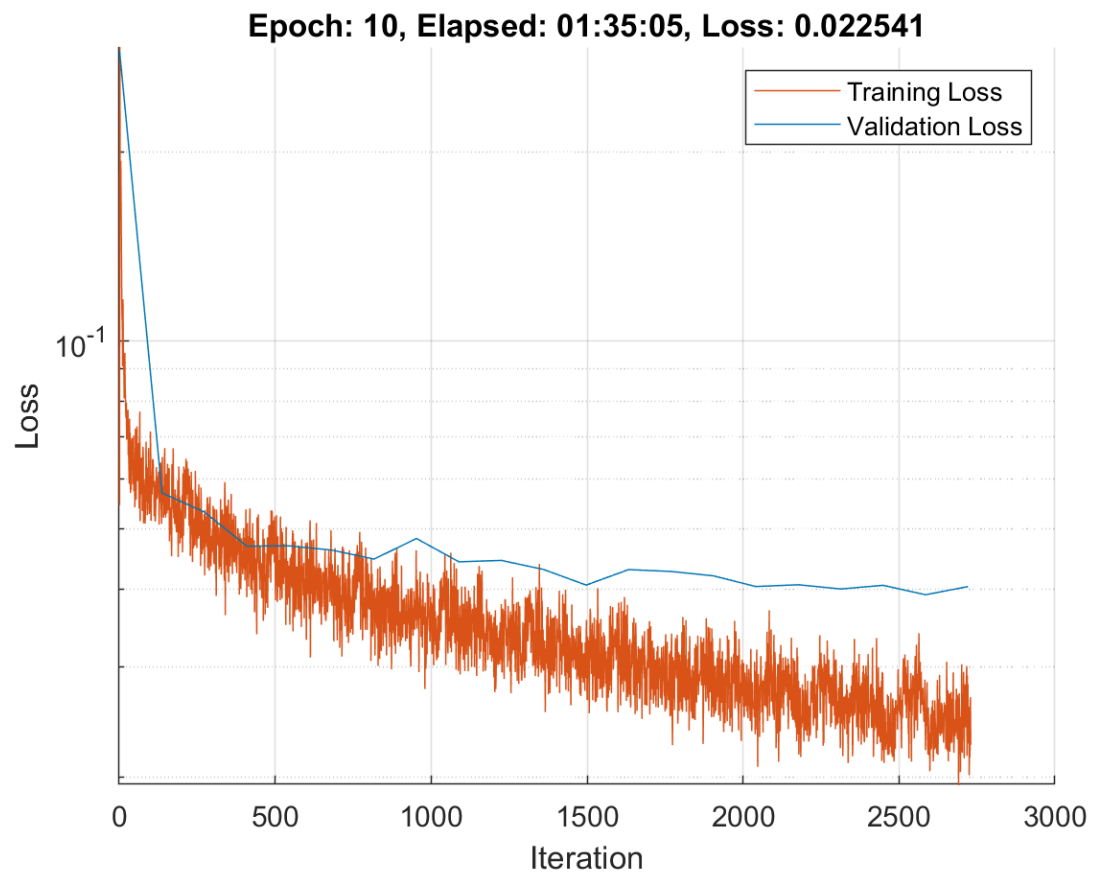


Figure 10: Training-Loss-Progression VGG TL, 5 Modes, 50,000 Datapoints

Task 6: Evaluation Of The VGG Networks Trained Using Transfer Learning (5 Modes)

The results are again visualized using a box plot of the correlation coefficient. Furthermore, the same metrics as in prior tasks have been listed in a table.

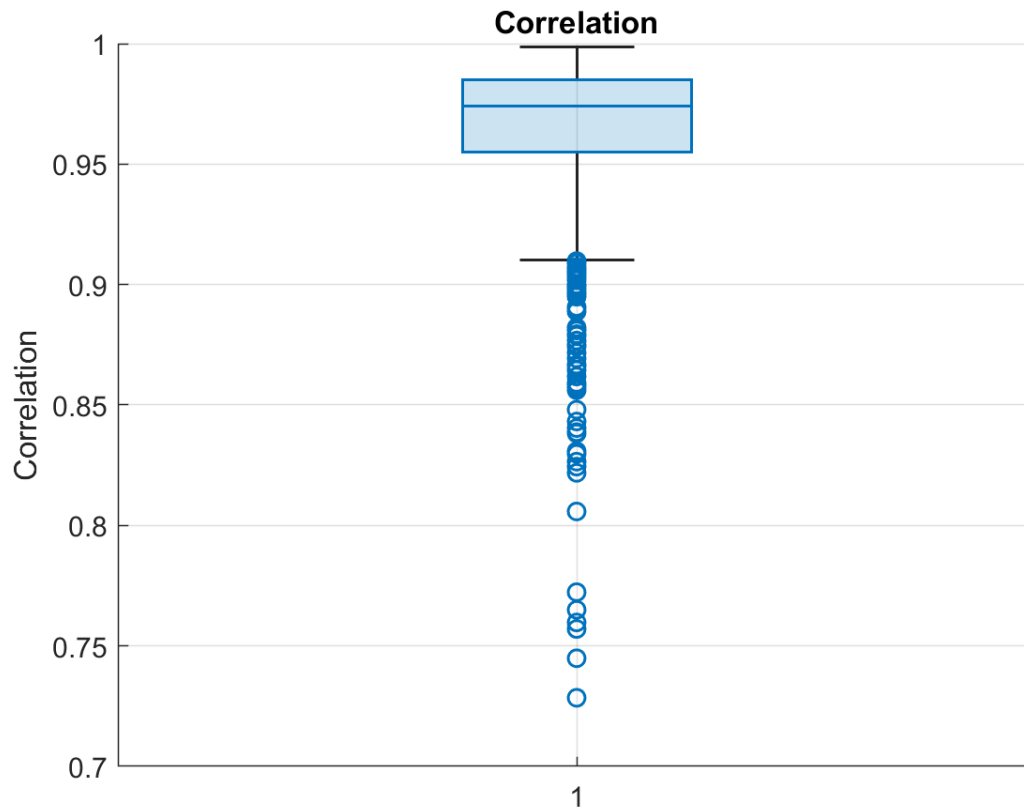


Figure 11: Evaluation VGG TL, 5 Modes, 10,000 Datapoints

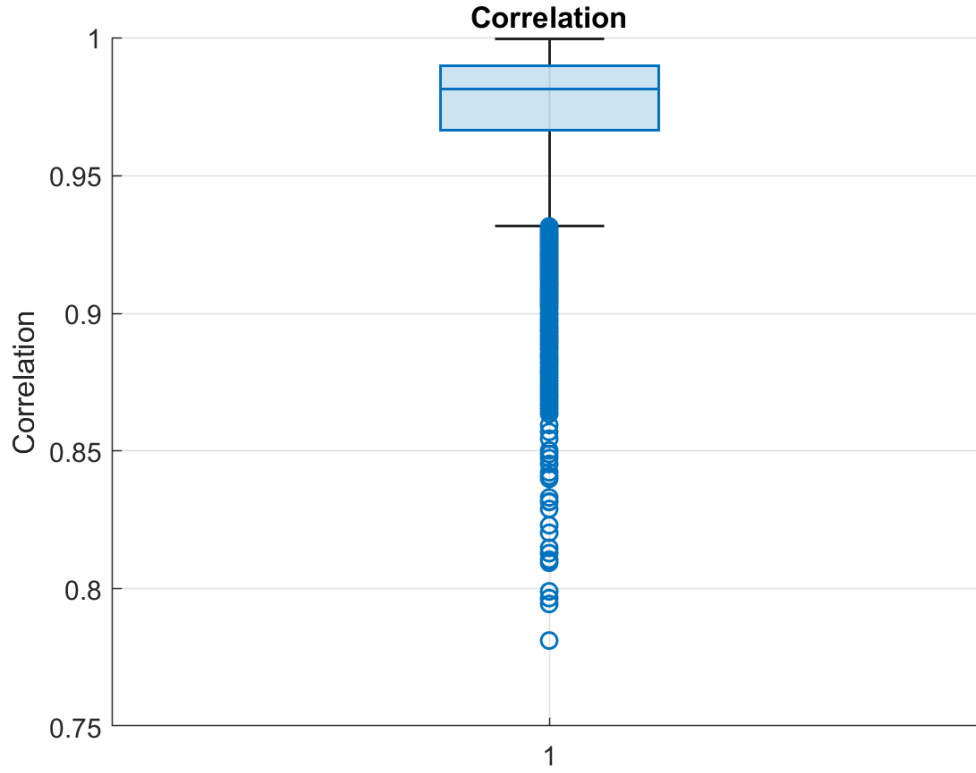


Figure 12: Evaluation VGG TL, 5 Modes, 50,000 Datapoints

VGG	Training data	$\bar{\Gamma}$	σ	$\bar{\Delta\rho}$	$\bar{\Delta\varphi}$
5-mode-TL	10000	96.5%	3.077%	9.4%	22.3%
5-mode-TL	50000	97.5%	2.216%	7.7%	16.4%

Task 7: Discussion

The results allow for a differentiated analysis of the performance of neural networks for mode compensation. Comparing the network architectures directly, shows that the VGG architecture is superior to a simple MLP. While an average correlation coefficient of 96.7% could be achieved using an MLP for 3 modes, the VGG architecture was able to reach 99.6% under identical conditions. This significant jump in performance is traceable to the convolutional layers of the VGG architecture, which can extract and process spatial characteristics as well as intensity distributions of the mode profiles in the input image far more effectively when compared to the flat architecture of the MLP.

Another aspect is the scaling of accuracy with the amount of investigated modes. When increasing the amount of modes from 3 to 5, every architecture and configuration showed predictive accuracy recessing. For the VGG architecture the average correlation coefficient fell from 99.6% to 96.4% whereas the MLP showed a slight decrease to 96.6%. This illustrates that the complexity of the reconstruction correlates with the number of coefficients to be predicted. The network must resolve a higher variance in transmission patterns as mode count increases. Strikingly, the relative deviation of the phase ($\overline{\Delta\varphi}$) was significantly higher than the one in amplitude ($\overline{\Delta\rho}$) for all experiments. This confirms the known physical challenge to precisely reconstruct phase information solely from intensity based image data without additional information.

The results from tasks 5 and 6 using transfer learning (TL) confirm that the size of the dataset used for training is critical for model stability and robustness. By using a network pretrained on 3 mode data, even 10,000 datapoints were able to yield solid results ($\overline{\Gamma} = 96.5\%$). An improvement to an average correlation of 97.5% as well as a significant reduction of the standard deviation σ to 2.216% was only achieved by expanding the training data to 50,000 datapoints. This demonstrates that transfer learning, while enabling efficient adaptation to more complex tasks, the resulting accuracy and robustness of the model depends primarily on training data volume.

In summary, it can be concluded that a CNN based architecture combined with transfer learning and a large amount of datapoints is most effective for accurate mode compensation.